

Improved lower bounds for the $3x + 1$ problem via the Krasikov–Lagarias inequalities at depths 3^{13} through 3^{20}

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Abstract

Let $\pi_1(x)$ denote the number of integers $n \leq x$ whose forward orbit under the $3x + 1$ map contains 1. Krasikov and Lagarias (2003) proved $\pi_1(x) \geq x^{0.84}$ for sufficiently large x by exhibiting feasible solutions of a linear-program family $L_k^{NT}(\lambda)$ at depth $k = 11$; this has remained the strongest unconditional lower bound (a December 2025 preprint, arXiv:2512.13760, still cites $x^{0.84}$ as the record). We solve $L_k^{NT}(\lambda)$ at depths $k = 13, 15, 17, 19, 20, 21$ (up to 3,486,784,401 constraints) and verify explicit rational certificates in exact integer arithmetic, obtaining

$$\pi_1(x) \geq x^{0.9184} \quad \text{for all sufficiently large } x,$$

with intermediate verified exponents 0.8624, 0.8805, 0.8953, 0.9069, 0.9146; the $k = 21$ exponent was pre-registered as a prediction ($\gamma \approx 0.918$, frozen 2026-07-16) before the computation. No new mathematics is required beyond [1]: Theorem 2.2 there converts any feasible solution into the density bound with exponent $\log_2 \lambda$. Our contribution is computational: a monotone (nonlinear Perron) iteration replaces the interior-point solver, making depths 3^{19} – 3^{20} feasible on a desktop machine (and 3^{20} classes at $k = 21$ via memmapped strictly-sequential sweeps on an external drive), and every certificate is re-verified with all weights replaced by strict rational lower bounds. We further observe an exact scalar identity governing the family (the “min-loss identity”), which predicts the feasibility edges to three decimals and suggests the method attains $x^{1-\varepsilon}$ for every $\varepsilon > 0$.

1 The system

Fix $k \geq 2$ and let $[3^k] = \{m \bmod 3^k : m \equiv 2 \bmod 3\}$, $\alpha = \log_2 3$. Following [1, §2], the linear program $L_k^{NT}(\lambda)$ has principal variables $c^m \geq 1$ for $m \in [3^k]$, auxiliaries $\bar{c}^t = \min(c^t, c^{t+3^{k-1}}, c^{t+2 \cdot 3^{k-1}})$ for $t \in [3^{k-1}]$, and constraints

$$m \equiv 2 \pmod{9} : \quad c^m \leq c^{4m} \lambda^{-2} + \bar{c}^{(4m-2)/3} \lambda^{\alpha-2}, \quad (1)$$

$$m \equiv 5 \pmod{9} : \quad c^m \leq c^{4m} \lambda^{-2}, \quad (2)$$

$$m \equiv 8 \pmod{9} : \quad c^m \leq c^{4m} \lambda^{-2} + \bar{c}^{(2m-1)/3} \lambda^{\alpha-1}, \quad (3)$$

with indices taken mod 3^k resp. 3^{k-1} . By [1, Thm. 2.2], a feasible positive solution at parameter λ yields $\pi_a(x) \geq \Delta_1 c^m \lambda^y$ for every $a \not\equiv 0 \bmod 3$, hence a density exponent $\gamma = \log_2 \lambda$.

*With extensive machine assistance (Claude/Jengo); all results independently verifiable from the deposited certificates.

2 Method and results

The constraint map F_λ (right-hand sides, with auxiliaries substituted) is monotone, positively homogeneous and concave; feasibility is equivalent to its nonlinear Perron root being ≥ 1 , located by power iteration and bisection in λ . The implementation reproduces the published values at $k = 9$ ($\gamma = 0.8168$; cf. 0.81 in [2]) and $k = 11$ ($\gamma = 0.8418$; cf. 0.84 in [1]). At greater depths:

k	classes	λ_0 (rational)	min. slack	verified exponent
13	531,441	1818/1000	1.000279	0.8624
15	4,782,969	1841/1000	1.000307	0.8805
17	43,046,721	93/50	1.000504	0.8953
19	387,420,489	15/8	1.000939	0.9069
20	1,162,261,467	1885/1000	1.000194	0.9146
21	3,486,784,401	1890/1000	1.000187	0.9184

Theorem 1. $\pi_1(x) \geq x^{0.9184}$ for all sufficiently large x .

Verification. For each depth the deposited integer vector C (scaled Perron vector) is checked against all constraints with λ_0 rational and every weight replaced by a strict rational lower bound (80-digit decimal floors minus one ulp, common denominator 10^{18}); comparisons are exact integer arithmetic. Zero violations at all six depths (5.1×10^9 constraints in total, of which 3.49×10^9 at $k = 21$ alone). Certificates and a standalone `numpy`-only verifier are deposited with this note. Three independent cross-checks strengthen the claim: (i) a second, independently written implementation (different indexing and λ parametrization) reproduces the feasibility edges at overlapping depths to 3–4 decimals ($k=13$: 0.8620; $k=15$: 0.8801; $k=17$: 0.8950); (ii) the $k = 12$ certificate ($\gamma = 213/250 = 0.852$, already above the previous record) has been machine-verified in *Lean 4*: the coefficient lower bounds reduce to integer power inequalities (e.g. $p_a^{250} \leq 2^{250Q-426}$) and all 177,147 class inequalities are checked by compiled kernel evaluation, with no floating point anywhere; (iii) both implementations were calibrated against all published anchor values ($k = 2, 9, 11$).

3 The min-loss identity and the ceiling

Summing the edge equalities over all classes (the maps $m \mapsto 4m$ and the two branch maps are class bijections) collapses the system to a scalar law: at the Perron edge,

$$1 = \lambda^{-2} + \frac{q}{3} (\lambda^{\alpha-2} + \lambda^{\alpha-1}), \quad q = \frac{3 \sum_t \bar{c}^t}{\sum_m c^m}.$$

(Verified to 10^{-8} at $k = 5, \dots, 11$.) Note $q = 1$ forces $\lambda = 2$, i.e. $\gamma = 1$: the entire gap between this method and full density is the “min-loss” $1 - q$. Measured q rises 0.888 ($k=5$) $\rightarrow$ 0.9705 ($k=19$), driven by equalization of the refinement triples of the Perron vector; the within-vector heterogeneity profile decays geometrically in the trit position with bulk factor ≈ 0.84 , matching the subleading spectral ratio of the edge-linearized operator. If this contraction admits a uniform bound below 1 (supported by all measurements; we leave it as a conjecture), then $q \rightarrow 1$ and the method attains $\pi_1(x) \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$, as hoped in [1].

Remark 1. *Depth $k = 21$ was completed after the first draft of this note: the pre-registered prediction $\gamma \approx 0.918$ (frozen before the computation) was confirmed at $\gamma = 0.9184$, on a desktop by memmapped strictly-sequential sweeps. Depths $k = 25\text{--}30$ (cloud scale) would discriminate between the extrapolations $\gamma_\infty \approx 0.95$ (ceiling) and $\gamma_\infty = 1$ (density); all flow-fraction measurements to date favor the latter.*

References

- [1] I. Krasikov, J. C. Lagarias, *Bounds for the $3x+1$ problem using difference inequalities*, Acta Arith. 109 (2003), 237–258; arXiv:math/0205002.
- [2] D. Applegate, J. C. Lagarias, *Density bounds for the $3x+1$ problem II: Krasikov inequalities*, Math. Comp. 64 (1995), 427–438.